Deepak Choudhary

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Professional Summary

Lead ML Engineer with over nine years of experience in search, ranking and NLP for e-commerce and job marketplaces. Deep expertise in hybrid retrieval, dense embeddings, learning-to-rank and offline relevance evaluation. Has led a team of five and recently introduced LLM-based query rewriting and reranking to production search.

Technical Skills

Programming: Python, Java, SQL

Search and Retrieval: Elasticsearch, OpenSearch, BM25, hybrid search, dense retrieval, cross-encoder re-ranking, query expansion

Embeddings and Vector Search: sentence-transformers, FAISS, Milvus, HNSW index tuning, embedding fine-tuning

Ranking and Evaluation: LambdaMART learning-to-rank, NDCG, MRR, recall at k, interleaving, A/B testing

LLM Applications: LLM query rewriting, LLM-based relevance labelling, prompt engineering

Engineering: Kafka, Spark, Docker, AWS, FastAPI, team leadership

Work Experience

Lead ML Engineer - Search and NLP - Hindon Marketplace Technologies Pvt Ltd

Noida, Uttar Pradesh | Sep 2022 - Present

- Lead a five-person team owning search relevance for a job marketplace with about 8 million documents.
- Introduced a hybrid retrieval stack combining BM25 and fine-tuned bi-encoder embeddings in Milvus with cross-encoder re-ranking; NDCG@10 improved by 14% in offline tests and 6% click-through in an A/B test.
- Created a relevance judgement pipeline using LLM labelling validated against 2,000 human labels (agreement 87%), enabling weekly offline evaluation.
- Introduced LLM-based query rewriting for long-tail queries, reducing zero-result searches by 31%.

ML Engineer - Hindon Marketplace Technologies Pvt Ltd

Noida, Uttar Pradesh | Apr 2020 - Aug 2022

- Built a LambdaMART ranker with 60 features and online feature logging.
- Developed an entity extraction model for skills and titles that powered filters and query understanding.
- Tuned HNSW parameters to balance recall and latency, reaching 95% recall at 25 ms.

Software Engineer - Search - Gomti Commerce Pvt Ltd

Noida, Uttar Pradesh | Aug 2017 - Mar 2020

- Maintained Elasticsearch clusters and implemented custom analyzers for multilingual product search.
- Built Kafka-based indexing pipelines handling about 2 million document updates per day.

Projects

Semantic Job-Candidate Matching

- Problem: Keyword search missed relevant candidates with differently worded skills.
- Approach: Fine-tuned a bi-encoder on click and application pairs with hard negatives and added a re-ranking stage.
- Technologies: PyTorch, sentence-transformers, Milvus
- Results: Recall at 50 rose from 0.62 to 0.78 on the evaluation set.

Search Relevance Evaluation Platform

- Problem: Relevance changes were tested only through slow online experiments.
- Approach: Built a platform that replays queries against candidate configurations and reports NDCG, MRR and failure examples.
- Technologies: Python, OpenSearch, Airflow
- Results: Cut experiment iteration time from two weeks to two days.

Education

M.Tech, Computer Science (part-time), Hindon Institute of Engineering, Noida (2018 - 2020) - CGPA 8.4 / 10

B.Tech, Computer Science and Engineering, Hindon Institute of Engineering, Noida (2013 - 2017) - CGPA 7.7 / 10

Certifications

- Elasticsearch Engineer Certification (Elastic)
- Natural Language Processing with Attention Models (Coursera, 2021)

Achievements

- Search relevance programme recognised with the company Innovation Award, 2023.
- Reviewer for an industry search-and-IR workshop, 2024.